

RTL WITH VERILOG LAB 3

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ROLL NO: 22FA-036-CE

Q1) Modified Traffic Light Controller

Design Code:

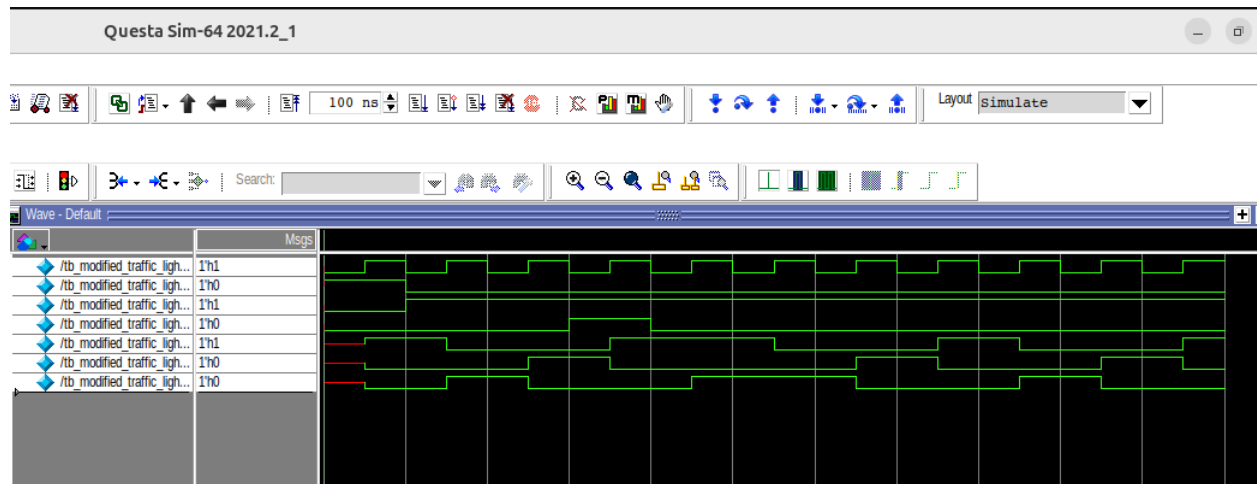
```
h /home/c4/RTL_WITH_VERILOG3/modified_traffic_light_controller.v - Default ;
Ln#
1  module modified_traffic_light_controller (
2      input clk,
3      input reset,
4      input button,
5      input ped_button,
6      output reg red,
7      output reg yellow,
8      output reg green
9  );
10 localparam RED_STATE = 2'b00,
11 GREEN_STATE = 2'b01,
12 YELLOW_STATE = 2'b10,
13 PED_STATE = 2'b11;
14 reg [1:0] current_state;
15 reg [1:0] next_state;
16 reg ped_request;
17 always @(posedge clk) begin
18     if (reset) begin
19         current_state <= RED_STATE;
20         ped_request <= 1'b0;
21     end
22     else begin
23         if (ped_button)
24             ped_request <= 1'b1;
25         if (button) begin
26             current_state <= next_state;
27             if (current_state == PED_STATE)
28                 ped_request <= 1'b0;
29         end
30     end
31 end
32 always @(*) begin
33     case (current_state)
34     RED_STATE: begin
35         if (ped_request)
36             next_state = PED_STATE;
37         else
38             next_state = GREEN_STATE;
```

```
39     end
40     GREEN_STATE:
41     next_state = YELLOW_STATE;
42     YELLOW_STATE:
43     next_state = RED_STATE;
44     PED_STATE:
45     next_state = GREEN_STATE;
46     default:
47     next_state = RED_STATE;
48     endcase
49     end
50     always @(*) begin
51     red = 1'b0;
52     yellow = 1'b0;
53     green = 1'b0;
54     case (current_state)
55     RED_STATE:
56     red = 1'b1;
57     GREEN_STATE:
58     green = 1'b1;
59     YELLOW_STATE:
60     yellow = 1'b1;
61     PED_STATE: begin
62     red = 1'b1;
63     green = 1'b1;
64     end
65     endcase
66     end
67     endmodule
68
```

Testbench Code:

```
h /home/c4/RTL_WITH_VERILOG3/tb_modified_traffic_light_controller.v - Default ;
Ln#
1  module tb_modified_traffic_light_controller;
2      reg clk;
3      reg reset;
4      reg button;
5      reg ped_button;
6      wire red;
7      wire yellow;
8      wire green;
9      modified_traffic_light_controller uut (
10         .clk(clk),
11         .reset(reset),
12         .button(button),
13         .ped_button(ped_button),
14         .red(red),
15         .yellow(yellow),
16         .green(green)
17     );
18     always #5 clk = ~clk;
19     initial begin
20         clk = 0;
21         reset = 1;
22         button = 0;
23         ped_button = 0;
24     $monitor("Time=%0t | Reset=%b Button=%b Ped_Button=%b | Red=%b Yellow=%b Green=%b
25     $time, reset, button, ped_button, red, yellow, green);
26     #10;
27     reset = 0;
28     button = 1; #20;
29     ped_button = 1; #10;
30     ped_button = 0; #30;
31     button = 1; #40;
32     $finish;
33     end
34 endmodule
```

QuestaSim Waveform:



FPGA Commands:

```
cl4@CL4-29: ~/f4pga-examples/xc7

cl4@CL4-29:~$ cd ~/f4pga-examples
cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/.opt/f4pga
cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
cl4@CL4-29:~/f4pga-examples$ source "$F4PGA_INSTALL_DIR/$FPGA_FAM/conda/etc/profile.d/conda.sh"
cl4@CL4-29:~/f4pga-examples$ conda activate xc7
(xc7) cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/.opt/f4pga
(xc7) cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
(xc7) cl4@CL4-29:~/f4pga-examples$ cd xc7
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ make -C mod_traffic clean
make: Entering directory '/home/cl4/f4pga-examples/xc7/mod_traffic'
/home/cl4/f4pga-examples/xc7/mod_traffic/../../common/common.mk:39: *** Unsupported board type. Stop.
make: Leaving directory '/home/cl4/f4pga-examples/xc7/mod_traffic'
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ TARGET="arty_100" make -C mod_traffic
make: Entering directory '/home/cl4/f4pga-examples/xc7/mod_traffic'
mkdir -p /home/cl4/f4pga-examples/xc7/mod_traffic/build/arty_100
cd /home/cl4/f4pga-examples/xc7/mod_traffic/build/arty_100 && symbiflow_synth -t modified_traffic_light_control
t_controller.v -d artix7 -p xc7a100tcs9324-1 -x /home/cl4/f4pga-examples/xc7/mod_traffic/arty.xdc
[F4PGA] Running (deprecated) synth
-t
modified_traffic_light_controller
-v
/home/cl4/f4pga-examples/xc7/mod_traffic/modified_traffic_light_controller.v
-d
artix7
```

Q2) Vending Machine

Design Code:

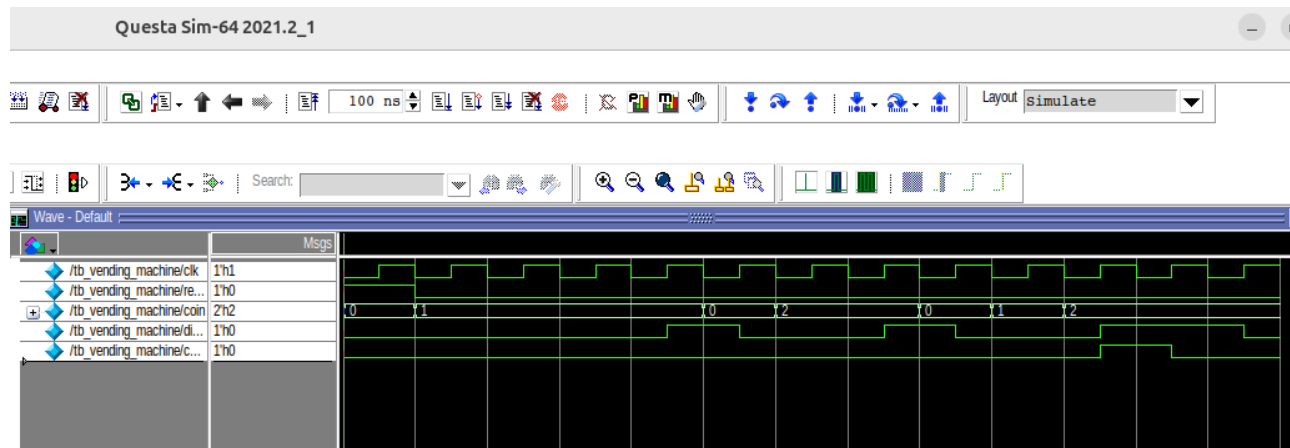
```
h /home/c4/RTL_WITH_VERILOG3/vending_machine.v - Default
Ln#
1 module vending_machine (
2     input clk,
3     input reset,
4     input [1:0] coin,
5     output reg dispense,
6     output reg change
7 );
8     localparam S0 = 3'b000,
9     S5 = 3'b001,
10    S10 = 3'b010,
11    S15 = 3'b011,
12    S20 = 3'b100;
13    reg [2:0] current_state;
14    reg [2:0] next_state;
15
16    always @(posedge clk or posedge reset) begin
17        if (reset)
18            current_state <= S0;
19        else
20            current_state <= next_state;
21    end
22
23    always @(*) begin
24        case (current_state)
25        S0: begin
26            if (coin == 2'b01)
27                next_state = S5;
28            else if (coin == 2'b10)
29                next_state = S10;
30            else
31                next_state = S0;
32        end
33        S5: begin
34            if (coin == 2'b01)
35                next_state = S10;
36            else if (coin == 2'b10)
37                next_state = S15;
38            else
```

```
39     next_state = S5;
40 end
41 S10: begin
42     if (coin == 2'b01)
43         next_state = S15;
44     else if (coin == 2'b10)
45         next_state = S20;
46     else
47         next_state = S10;
48     end
49 S15: begin
50     if (coin == 2'b01)
51         next_state = S20;
52     else if (coin == 2'b10)
53         next_state = S20;
54     else
55         next_state = S15;
56     end
57 S20: next_state = S0;
58 default: next_state = S0;
59 endcase
60 end
61
62 always @(*) begin
63     dispense = 1'b0;
64     change = 1'b0;
65 case (current_state)
66 S15: begin
67     if (coin == 2'b10) begin
68         dispense = 1'b1;
69         change = 1'b1;
70     end
71     end
72 S20: begin
73     dispense = 1'b1;
74     end
75 endcase
76 end
77 endmodule
```

Testbench Code:

```
1  module tb_vending_machine;
2      reg clk;
3      reg reset;
4      reg [1:0] coin;
5      wire dispense;
6      wire change;
7
8  □ vending_machine uut (
9      .clk(clk),
10     .reset(reset),
11     .coin(coin),
12     .dispense(dispense),
13     .change(change)
14 );
15
16     always #5 clk = ~clk;
17 □ initial begin
18     clk = 0;
19     reset = 1;
20     coin = 2'b00;
21 □ $monitor("Time=%0t | Reset=%b Coin=%b | State=%d Dispense=%b Change=%b",
22     $time, reset, coin, uut.current_state, dispense, change);
23     #10;
24     reset = 0;
25     coin = 2'b01; #10;
26     coin = 2'b01; #10;
27     coin = 2'b01; #10;
28     coin = 2'b01; #10;
29     coin = 2'b00; #10;
30     coin = 2'b10; #10;
31     coin = 2'b10; #10;
32     coin = 2'b00; #10;
33     coin = 2'b01; #10;
34     coin = 2'b10; #10;
35     coin = 2'b10; #20;
36     $finish;
37     end
38 endmodule
```

QuestaSim Waveform:



FPGA Commands:

```
cl4@CL4-29: ~/f4pga-examples/xc7
cl4@CL4-29:~$ cd ~/f4pga-examples
cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/.opt/f4pga
cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
cl4@CL4-29:~/f4pga-examples$ source "$F4PGA_INSTALL_DIR/$FPGA_FAM/conda/etc/profile.d/conda.sh"
cl4@CL4-29:~/f4pga-examples$ conda activate xc7
(xc7) cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/.opt/f4pga
(xc7) cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
(xc7) cl4@CL4-29:~/f4pga-examples$ cd xc7
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ make -C vending clean
make: Entering directory '/home/cl4/f4pga-examples/xc7/vending'
/home/cl4/f4pga-examples/xc7/vending/../../common/common.mk:39: *** Unsupported board type. Stop.
make: Leaving directory '/home/cl4/f4pga-examples/xc7/vending'
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ TARGET="arty_100" make -C vending
make: Entering directory '/home/cl4/f4pga-examples/xc7/vending'
mkdir -p /home/cl4/f4pga-examples/xc7/vending/build/arty_100
cd /home/cl4/f4pga-examples/xc7/vending/build/arty_100 && symbiflow_synth -t vending_machine -v /home/cl4/f4pg
-x /home/cl4/f4pga-examples/xc7/vending/arty.xdc
[F4PGA] Running (deprecated) synth
-t
vending_machine
-v
/home/cl4/f4pga-examples/xc7/vending/vending_machine.v
-d
artix7
```


Q3) Modified Synchronous FIFO

Design Code:

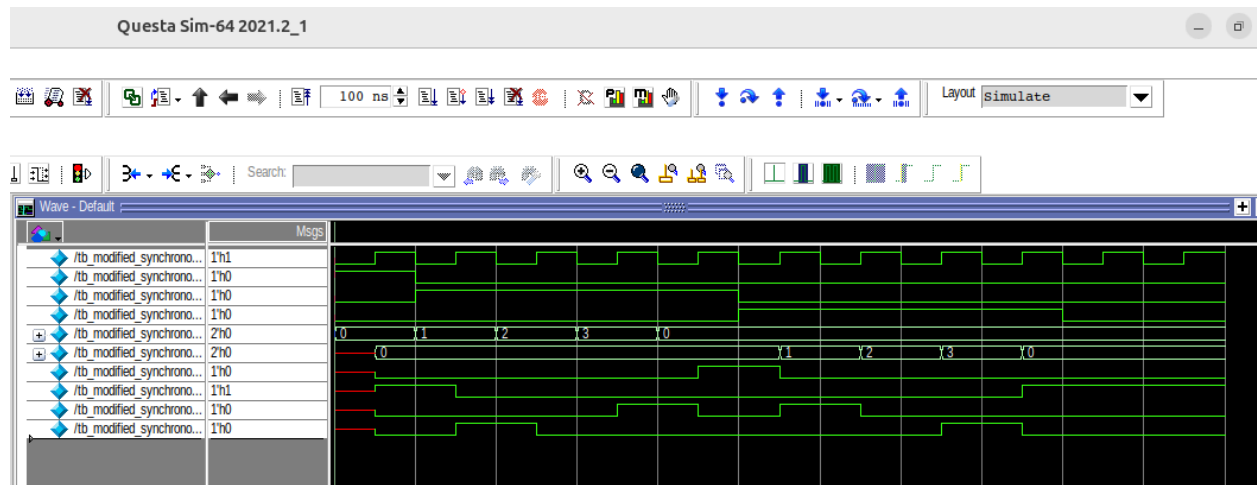
```
h /home/cl4/RTL_WITH_VERILOG3/modified_synchronous_fifo.v - Default
Ln#
1  module modified_synchronous_fifo (
2      input clk,
3      input reset,
4      input write_enable,
5      input read_enable,
6      input [1:0] data_in,
7      output reg [1:0] data_out,
8      output full,
9      output empty,
10     output almost_full,
11     output almost_empty
12 );
13     parameter DEPTH = 4;
14     reg [1:0] memory [0:DEPTH-1];
15     reg [1:0] write_pointer;
16     reg [1:0] read_pointer;
17     reg [2:0] count;
18
19     always @(posedge clk) begin
20     if (reset) begin
21         write_pointer <= 2'd0;
22         read_pointer <= 2'd0;
23         count <= 3'd0;
24         data_out <= 2'd0;
25     end
26     else begin
27     case ({write_enable && !full, read_enable && !empty})
28     2'b10: begin
29         memory[write_pointer] <= data_in;
30         write_pointer <= write_pointer + 1'b1;
31         count <= count + 1'b1;
32     end
33     2'b01: begin
34         data_out <= memory[read_pointer];
35         read_pointer <= read_pointer + 1'b1;
36         count <= count - 1'b1;
37     end
38     endcase
39     end
40 end
```

```
38 2'b11: begin
39     memory[write_pointer] <= data_in;
40     data_out <= memory[read_pointer];
41     write_pointer <= write_pointer + 1'b1;
42     read_pointer <= read_pointer + 1'b1;
43 end
44 default: ;
45 endcase
46 end
47 end
48
49 assign full = (count == DEPTH);
50 assign empty = (count == 0);
51 assign almost_full = (count == DEPTH - 1);
52 assign almost_empty = (count == 1);
53 endmodule
54
```

Testbench Code:

```
h /home/cl4RTL_WITH_VERILOG3/tb_modified_synchronous_fifo.v - Default
Ln#
1 module tb_modified_synchronous_fifo;
2   reg clk;
3   reg reset;
4   reg write_enable;
5   reg read_enable;
6   reg [1:0] data_in;
7   wire [1:0] data_out;
8   wire full;
9   wire empty;
10  wire almost_full;
11  wire almost_empty;
12
13  modified_synchronous_fifo uut (
14    .clk(clk),
15    .reset(reset),
16    .write_enable(write_enable),
17    .read_enable(read_enable),
18    .data_in(data_in),
19    .data_out(data_out),
20    .full(full),
21    .empty(empty),
22    .almost_full(almost_full),
23    .almost_empty(almost_empty)
24  );
25
26  always #5 clk = ~clk;
27  initial begin
28    clk = 0;
29    reset = 1;
30    write_enable = 0;
31    read_enable = 0;
32    data_in = 2'd0;
33    $monitor("Time=%0t | WE=%b RE=%b DataIn=%d DataOut=%d | Full=%b AF=%b AE=%b Empty=%b",
34      $time, write_enable, read_enable, data_in, data_out, full, almost_full, almost_empty, empty);
35    #10;
36    reset = 0;
37    write_enable = 1;
38
39    data_in = 2'd1; #10;
40    data_in = 2'd2; #10;
41    data_in = 2'd3; #10;
42    data_in = 2'd0; #10;
43    write_enable = 0;
44    read_enable = 1; #10;
45    read_enable = 1; #10;
46    read_enable = 1; #10;
47    read_enable = 1; #10;
48    read_enable = 0; #20;
49    $finish;
50  end
endmodule
```

QuestaSim Waveform:



FPGA Commands:

```
cl4@CL4-29: ~/f4pga-examples/xc7
cl4@CL4-29:~$ cd ~/f4pga-examples
cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/.opt/f4pga
cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
cl4@CL4-29:~/f4pga-examples$ source "$F4PGA_INSTALL_DIR/$FPGA_FAM/conda/etc/profile.d/conda.sh"
cl4@CL4-29:~/f4pga-examples$ conda activate xc7
(xc7) cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/.opt/f4pga
(xc7) cl4@CL4-29:~/f4pga-examples$ cd ~/f4pga-examples
(xc7) cl4@CL4-29:~/f4pga-examples$ cd xc7
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ make -C mod_fifo clean
make: Entering directory '/home/cl4/f4pga-examples/xc7/mod_fifo'
/home/cl4/f4pga-examples/xc7/mod_fifo/../../common/common.mk:39: *** Unsupported board type. Stop.
make: Leaving directory '/home/cl4/f4pga-examples/xc7/mod_fifo'
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ TARGET="arty_100" make -C mod_fifo
make: Entering directory '/home/cl4/f4pga-examples/xc7/mod_fifo'
mkdir -p /home/cl4/f4pga-examples/xc7/mod_fifo/build/arty_100
cd /home/cl4/f4pga-examples/xc7/mod_fifo/build/arty_100 && symbiflow_synth -t modified_synchronous_fifo -v /ho
x7 -p xc7a100tcs324-1 -x /home/cl4/f4pga-examples/xc7/mod_fifo/arty.xdc
[F4PGA] Running (deprecated) synth
-t
modified_synchronous_fifo
-v
/home/cl4/f4pga-examples/xc7/mod_fifo/build/arty_100/modified_synchronous_fifo.v
-d
artix7
```